

Wood Heat and Hot Water

Carpathian OER Commons — Utilities Module

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Maturity: L0, a marked draft — a sizing method and a design, published so that they can be read and changed in the open. The buildings the heat serves have not been surveyed, so no heat load, boiler size, or year's firewood is given for the site; the priced lines are the working papers' ranges for autumn 2026, and the chimneys, the woodshed, the firewood, and the safety fittings are not priced. Read every figure below at this level and no higher. The module moves to L1 when the buildings are surveyed, the heat load is calculated for each, and every line in Section 6 carries a dated cost and its basis.

Governed by Section 2 of the compendium. No child appears in this module in any form. The site is not named; site plans are schematic; photographs show structures and systems only. Design loads are site capacity, used for sizing only.

Converted from the energy working paper, EN v0.1, Section 8, and the sanitation working paper, EN v0.2, Sections 7 and 8.7, the water heater and the solar collectors. Five things change from the papers, decided on 2026-10-03: every wood fire on the site is in this module, the cookstove and the masonry heater as appliances, with the chimneys and the firewood for all of them; the buffer tank is sized by rule, and carries the sleeping rooms' heat after the fire rules put the fire out two hours before sleep; the solar loop is pumped, never a water-filled thermosiphon; firewood is bought and priced, never counted as the household's contribution of labour, and the site's own wood enters only on a felling ticket, by paid hands; and gravity circulation is preferred, so that the heat does not wait on electricity.

Permits and local rules. This module is a guide, used in different regions and countries under different laws and guidelines. Depending on where it is built, at what scale, and for whom, permits, registrations, or approvals may be required — for building, water, sanitation, fire, waste, food, or the use of forest and land. Section 10 records what was found for Ukraine, dated, and unchecked until signed; a reader elsewhere checks their own rules with the competent authorities before building, and nothing here replaces that.

1. Purpose

Warmth for the sleeping accommodation and the shared rooms, and hot water for the washroom and the kitchen, from wood and the sun, through a mountain winter and through power cuts. A log boiler into a buffer tank, radiators fed by gravity, a masonry heater in the gathering place, a cookstove in the kitchen, a small water heater and solar collectors for the summer, and a year's firewood dried under a roof.

Six rules. No heat from electricity: wood and the sun carry the heat, and the energy module carries none of it. Burn hot, store the heat: a log boiler burns at full output into a buffer tank, never throttled, because a smouldering fire makes tar and smoke. The fire out before sleep, the warmth kept: where people stay round the clock, firing ends two hours before sleep, and the buffer and the masonry heater carry the night. Gravity before pumps: heat moves from boiler to buffer to radiators by gravity where the

buildings allow it. Dry wood only: under 20 percent moisture, split, a year ahead. Bought or felled by right: firewood is bought and priced, or felled on a ticket by paid hands; the deadwood stays in the forest, and no heat depends on any work the programme does.

Connections. Depends on: the buildings modules — sleeping accommodation, the kitchen and dining, the covered gathering place and hearth, insulation — for the heat load of each building, and the kitchen and the gathering place for whether and where a fire stands in their rooms; the Phase 1 building survey; the water module, for cold water and its pressure; the energy module, for the solar loop's pump and any controls; the written enquiries in Section 10. Feeds: the washroom block, with hot water from the tank outlet on; the kitchen, with hot water and its cookstove; the sleeping accommodation and the shared rooms, with heat. Takes from: the micro-hydro module's dump load, as an option.

Patterns. Grows from: not yet named. How the stove stood to the house, whether an open fire belongs in the gathering place beside a masonry heater or instead of it, where the woodpile stands, and which wood is not burned are questions for the elders; the literature records a cooking ledge with an open fire beside the stove, and a firewood shed against the house [for the elders]. Feeds: not yet named. Resonates with: 230 Radiant Heat — the masonry heater and the radiators warm by radiation, with a seat against the heater's warm side; 145 Bulk Storage — a year's wood kept dry under a roof, ahead of its burning; 128 Indoor Sunlight — every room on the south asks less wood; 83 Master and Apprentices — paid hands beside the stove-builder. In tension with: 181 The Fire — the fire rules end firing two hours before sleep where people stay round the clock, and a boiler in its own room hides its fire; the masonry heater keeps a fire at the centre by day and its heat by night. 7 The Countryside and 172 Garden Growing Wild — wood is taken only where tenure and a felling ticket allow, and the deadwood stays. 105 South Facing Outdoors — the solar collectors share the south roof with the array. The walk status of each is set after the Phase 1 walk, on a later date; until then, unknown (compendium Section 26).

2. Context of validity

Condition	Range this design holds for
Climate	design temperature -20°C ; 175–250 heating days
Altitude	[set]
Buildings	surveyed; heat load per building, [set]
Group size	capacity: 34 people, for hot water
Season of use	all seasons; heat in the heating season
Fuel	firewood, bought or felled on a ticket

Condition	Range this design holds for
Power	none to heat by gravity; the solar loop's pump
Water	the water module's supply and pressure

The design temperature and the heating season are the building-climate standard's for the oblast stations of the Carpathians (ДСТУ-Н Б В.1.1-27:2010, Table 2): the coldest five days -19 to -20 °C, and -15 to -20 in Zakarpattia, and 175 to 250 days with a daily mean below 8 °C, the longest at the mountain stations. A site higher than the stations is colder.

3. Sizing method

What you must measure yourself before using this module:

- **Your buildings** — each one's heated floor, walls, roof, windows, and how it is heated now, from a survey; its heat loss at the design temperature, calculated by EN 12831 or by an engineer.
- **Your hot water** — the hot side's meter over a full day at capacity, and the temperature of the cold water at the tap.
- **Your wood** — what is bought, by the stacked cubic metre, and its moisture, with a moisture meter on a freshly split face.
- **Your roof for the collectors** — its pitch, its aspect, and the share the array needs.

Our example: design_temperature, -20 °C; heating_season, 175 to 250 days; capacity, 34 people; hot_water_per_person, 40 L a day at 45 °C, [assumed]; use_days_per_year, 120; boiler_efficiency, 0.74 to 0.81; firewood_energy, 1,333 to 1,885 kWh a stacked m³; buffer_per_kw, 55 to 100 L a kW; exchange_rate, 52 UAH to the euro. Values from the site parameters file, held in the workshop.

3.1 The heat load

The heat load of a building is its heat loss at the design temperature, in kilowatts. Before a survey it can only be bracketed: a German rule-of-thumb table gives 157 to 163 W a m² of heated floor for an uninsulated building from before 1977, falling to 36 to 38 W a m² for one built since 2009 (Reonic, 2026), at German design temperatures that are milder than -20 °C. That is 3.6 to 16.3 kW for each 100 m² of heated floor, a plausibility check and no more; the survey and the calculation replace it, building by building, [set].

A new or renovated building here is held to ДБН В.2.6-31:2021, in force since 01.09.2022, which sets a thermal resistance of at least 4.0 m²·K a W for external walls in temperature zone I; the Carpathian oblasts are given as zone I by secondary sources, to be confirmed on the standard's own map. The insulation module carries it.

3.2 The firewood

Quantity	Calculation	Each 10,000 kWh of heat
Beech, stacked m ³	$10,000 \div (1,885 \times 0.74 - 0.81)$	about 6.5–7.2
Spruce, stacked m ³	$10,000 \div (1,333 \times 0.74 - 0.81)$	about 9.3–10.1

The heat a year is each building's heat load, its hours of heating, and its use, from the survey; until then the firewood is given per 10,000 kWh of heat delivered. The energy figures are at 20 percent water content, with 0.7 solid cubic metres to the stacked (LWF Merkblatt 12, 2024); Ukrainian sales count the складометр, and ГОСТ 3243-88 gives 0.63 to 0.72 solid to stacked by kind and split. The efficiency is what log boilers gave in use, 74 percent gross and 81 net, about fifteen points below the laboratory (Kiwa, for BEIS, 2019).

Drying. Split and covered, wood outdoors reached 20 percent in about six months and about 15 after nine; in poorer conditions it took a year or more, and unsplit wood longer (LWF and TFZ, drying trials, 2006). Stored wood loses up to about 3 percent of its heat a year, and wood left open re-wets to near 30 percent in autumn. So a year's wood is bought or felled a year ahead, split, and kept under a roof that covers its top, off the ground, with the wind through it.

3.3 The boiler and the buffer

Quantity	Calculation	Each 10 kW of boiler
Buffer, the legal minimum	$55 \text{ L} \times \text{kW}$	550 L
Buffer, one fill a day	$\text{about } 100 \text{ L} \times \text{kW}$	about 1,000 L
Buffer, by fuel chamber	$12 \text{ L} \times \text{the chamber's litres}$	[set]
Heat held, 85 to 60 °C	$1,000 \text{ L} \times 1.163 \text{ Wh} \times 25 \text{ K}$	about 29 kWh

The boiler. A log gasification boiler, sized to the heat load of the buildings it serves at the design temperature, [set]. It burns at full output into the buffer and is never throttled to keep the fire in: a buffer too small makes the boiler choke its air, and tar forms in the fuel chamber (Koop, haustec, 2022). Each start and stop cost the equivalent of one to five hours of steady running in the field trial (Kiwa, for BEIS, 2019). The return water is held above 55 to 60 °C by a thermal mixing valve, or tar and condensate corrode the boiler.

The buffer. The German rule for a hand-fed boiler is at least 12 L of buffer for each litre of fuel chamber and never less than 55 L a kW (1. BImSchV, §5, as given in BDH Infoblatt 22, 2011); for one fill a day, about 100 L a kW (Bavarian LfU). One boiler of this kind, 15 kW with an 80 L chamber, asks 1,000 L (Fröling, S1 Turbo data sheet).

The night. The fire rules end firing two hours before sleep where people stay round the clock (Section 10). The buffer carries the sleeping rooms through the night: a building that loses P kilowatts at the night's temperature asks $P \times$ the night's hours in kilowatt-hours, and each 1,000 L of buffer between 85 and 60 °C holds about 29 kWh, the range gravity radiators can use. The buffer is sized for the larger of the boiler's rule and the night's need, [set] from the survey.

3.4 Gravity circulation

Gravity moves the water because hot water is lighter: about 6.4 mbar of driving pressure for 5 m of height between the boiler's middle and the radiators' at 90 and 70 °C (Wikipedia, Schwerkraftheizung). The installer guides found ask for the radiators about 2 m or more above the boiler, mains of no more than about 25 m to the furthest riser, a slope of at least 5 mm a metre toward the boiler, and pipes much larger than a pumped system's (Klimosz; Ukrainian installer guides). The open expansion vessel stands at the highest point, at least 0.7 m above the highest radiator, with no valve of any kind on its safety and expansion pipes, the safety pipe at least $8.08 \times \sqrt{\text{kW}}$ mm and never under 25 mm, all insulated where the room can fall to 0 °C (PN-B-02413:1991, as taught at Wrocław University of Science and Technology).

Where a building does not allow it, the radiators are pumped, and the boiler keeps a gravity heat leak — a radiator on a pipe with no pump, for about a quarter of the boiler's output (Solid Fuel Association) — so that a power cut never leaves a full firebox with nowhere to put its heat. A thermal discharge valve, which opens at about 95 °C and runs cold water through a coil in the boiler to the drain, needs mains pressure to work (Watts, STS data sheet); the water module's gravity supply gives about 1 to 2 bar, so whether it is enough is checked with the valve's maker.

3.5 Hot water

Quantity	Calculation	Our example
A day at capacity	$34 \times 40 \text{ L at } 45 \text{ °C}$	1,360 L
Heat a kelvin of rise	$1,360 \text{ L} \times 1.163 \text{ Wh}$	about 1.58 kWh
From cold water at 8 °C, [assumed]	$\times 37 \text{ K}$	about 58.5 kWh
A year, on use days	$58.5 \text{ kWh} \times 120$	about 7,000 kWh
Beech, all by wood	$7,000 \div (1,885 \times 0.74\text{--}0.81)$	4.6–5.0 m ³
The paper's tank, 200 L at 60 °C	mixed to 45 °C	about 281 L

The hot water. The papers' 200 L tank holds about a fifth of a full day at capacity, so it is not a store but a draw-off, reheated from the buffer through a coil, with the sanitation paper's wood-fired water heater for the summer, when the boiler is cold. Hot water is held at 55 °C or more where the standard applies (ДБН В.2.5-64:2012, with Amendment 1), and mixed down at the tap.

The sun. Solar collectors give about 350 to 600 kWh a m² a year by type, flat plate the lower and evacuated tube the higher (Carbon Trust, for BEIS, 2020; solaranlage-ratgeber, 2026) — about 1.0 to 1.6 kWh a m² a day across the year, more in summer. Two collectors, their area [set] from the chosen collector, carry a share of the summer's hot water, [set]. The loop runs on about 40 percent propylene glycol, good to about -20 °C, checked every two years and replaced every five to ten (energie-experten, 2020), or as a drainback loop that empties when the pump stops. A water-filled passive thermosiphon is not used: it was found safe only in the warmest parts of the United States, and its freeze valves failed about a quarter of the time in hard frosts (Burch and Salasovich, NREL, 2005). The solar heat enters the buffer's lowest zone; the boiler never heats the middle, where it would block the sun's heat (Solarserver, 2025).

3.6 The masonry heater and the cookstove

The masonry heater. For the gathering place: at least 800 kg of mass, the smoke turned at least once through 180°, a gas path at least twice the firebox's largest dimension, walls averaging no more than 250 mm, the outside no hotter than 110 °C (Masonry Heater Association, 1998), fired hot once or twice a day and giving heat for about 12 to 24 hours (makers' figures, secondary). EN 15250 has been replaced by EN 16510-2-5:2025, which the maker's declaration names.

The cookstove. For the kitchen, if the kitchen module and its rules allow cooking on wood; its chimney is cleaned monthly under the fire rules.

Carbon monoxide. No Ukrainian rule was found that asks for carbon monoxide alarms; the design fits them on every floor where a fire burns and outside every sleeping room, as the US EPA advises (Burn Wise).

4. Bill of materials

Quantities for one building and the shared appliances; the survey multiplies them. Alternatives are named in the design; none is yet known to work here.

Item	Quantity	Alternatives in the design
Log gasification boiler	1, [set] kW	a simpler log boiler
Buffer tank, stratified, with coil	[set] L	two tanks

Item	Quantity	Alternatives in the design
Return-temperature mixing valve	1	a loading unit
Radiators and pipework, gravity	[set]	pumped, with a heat leak
Open expansion vessel, safety pipes	1 set	none named
Thermal discharge valve	1	none named
Masonry heater, 800 kg or more	1	none named
Wood cookstove	1	none named
Wood-fired water heater	1	none named
Solar collectors	2	evacuated tube
Hot water tank, 200 L	1	heat exchanger
Solar loop, pump, glycol	1 set	drainback
Chimneys, one for each fire	[set]	none named
Spark arresters	where roofs burn	none named
Carbon monoxide alarms	[set]	none named
Woodshed, roofed, open-sided	a year's wood	none named

The fires, as the rules ask. Every stove and the boiler stand at the clearances to combustible structures that ДБН В.2.5-67:2013 sets, with a metal sheet of 0.7 by 0.5 m on a wooden floor in front of each firebox; goods kept 0.7 m from a stove and 1.25 m from a firebox opening; chimneys plastered and whitewashed in the attic; spark arresters on chimneys through a roof that can burn (Fire Safety Rules, Section IV, chapter 2). No temporary stove anywhere.

The woodshed. Roofed, open to the wind, the wood on rails or pallets off the ground, a year's wood ahead; the fire distances from the buildings it serves are asked, [set].

The layout, as designed. Woodshed, boiler, buffer, then the radiators of each building by gravity; the solar collectors into the buffer's lowest zone; the tank and the water heater to the washroom and the kitchen; the masonry heater and the cookstove in their own rooms, each with its chimney. Drawn as relationships only.

5. Build

Step	Trade	Time	If done out of order
Buildings surveyed, loads	engineer	[from L2]	mis-sized
Enquiries answered	household	[from L2]	[from L2]
Chimneys	stove-builder	[from L2]	no draught
Boiler, buffer, valves	heating fitter	[from L2]	tar
Gravity pipework, vessel	heating fitter	[from L2]	no circulation
Masonry heater	stove-builder	[from L2]	cracks
Cookstove, water heater	stove-builder	[from L2]	[from L2]
Solar loop	heating fitter	[from L2]	frozen loop
Woodshed, a year's wood	builder	[from L2]	wet wood
First firing, chimney checked	stove-builder	[from L2]	[from L2]

6. Cost

Estimates, not quotations: every priced line is a range from the working papers for autumn 2026, marked "paper"; re-pricing from dated listings was tried on 2026-10-03 and could not be done. Place: not a quotation. Exchange rate: exchange_rate, 52 UAH to the euro, the National Bank's mid-September 2026 rate, rounded.

Item	UAH	EUR	Basis
Wood cookstove	30,000–45,000	577–865	paper, E 8
Masonry heater	80,000–120,000	1,538–2,308	paper, E 8
Boiler, buffer, radiators	150,000–250,000	2,885–4,808	paper
Wood-fired water heater	20,000	385	paper, S 8.7
Solar, 2 collectors, 200 L tank	45,000	865	paper, S 8.7
Chimneys	[set]	[set]	not priced

Item	UAH	EUR	Basis
Safety fittings and alarms	[set]	[set]	not priced
Woodshed	[set]	[set]	not priced
Subtotal, priced lines	325,000–480,000	6,250–9,231	

The boiler's line, from the energy paper, is for one building, and the survey says how many; no percentages are taken on a subtotal this incomplete. The papers budget the heat under the living layer; here it is counted as its own module.

Contribution, not in the total. Wood from the household's own woodland, where a felling ticket allows it, shown at the price of the same wood bought; the felling, splitting, and carrying are paid and priced. Nothing else.

Running cost.

- Firewood: the year's stacked cubic metres, from Section 3.2, at a dated local price, [set].
- Chimneys cleaned at the intervals in Section 10, by a sweep, [set].
- The solar loop's glycol checked every two years and replaced every five to ten, [set].
- The work is paid, [set] hours.

7. Operation and maintenance

Responsible: one named adult, trained in firing and in the fire rules, paid, with a deputy — [from L2]. Firing is adult work only, and no stove is left unattended.

- **Daily, in the heating season:** the boiler fired hot to fill the buffer, and firing ended two hours before sleep; the buffer's top and bottom temperatures read; the masonry heater fired once or twice.
- **Before each heating season:** every stove, the boiler, and every chimney checked and the result written down; the expansion vessel's water level read; the alarms tested.
- **Monthly:** the cookstove's chimney cleaned.
- **Every two or three months:** the other chimneys cleaned, as the fire rules set.
- **Spring:** the next winter's wood bought or felled, split, and stacked under the roof.
- **Every two years:** the solar glycol checked.

How the programme relates to this work is described in the programme modules, in standing format.

8. Failure modes

From the design, not from experience.

Failure	Early sign	Repair and cost
Tar in the boiler	black deposits	larger buffer; return valve
Wet wood	hissing, smoke, soot	dry wood a year ahead
Chimney fire	roaring, sparks	the fire service; sweep
Power cut, pumped loop	radiators cool	gravity heat leak
Gravity loop stalls	cold radiators	check level, slope, air
Solar loop frozen	no gain in sun	glycol strength checked
Buffer empty by morning	cold rooms at dawn	larger buffer
Carbon monoxide	alarm	ventilate; [from L2]

9. What we got wrong

Nothing yet: this module has not been built.

10. Regulatory notes

Country	Instrument, number, date	Checked by, date
Ukraine	found 2026-10-03, not checked — see below	—
Romania	not checked	—
Poland	not checked	—
Slovakia	not checked	—
Hungary	not checked	—
Serbia	not checked	—
Czechia	not checked	—

Ukraine, found and not checked. Read on 2026-10-03 on secondary copies — LIGA:ZAKON, protocol.ua, kodeksy.com.ua, and a university copy of the fire rules without their later amendments — because the statute site could not be reached; each is read on zakon.rada.gov.ua and signed before this row counts as checked.

- **Fire Safety Rules of Ukraine, Order No. 1417 of the Ministry of Internal Affairs, 30.12.2014**, amended last by Order No. 67 of 27.01.2026, Section IV, chapter 2: stoves and boilers fired by trained people; firing ended two hours before sleep in round-the-clock establishments; chimneys cleaned at least every three months for intermittent stoves, every two for continuous ones, and monthly for kitchen ranges; clearances to combustible structures as ДБН В.2.5-67:2013 sets, a metal sheet of 0.7 by 0.5 m before a firebox on a wooden floor, goods 0.7 m from a stove and 1.25 m from a firebox opening; spark arresters on combustible roofs; ash doused and taken to its place; no temporary stoves; never a stove unattended or minded by small children, never fired during mass events, never with wood too long for the firebox.
- **Fire Safety Rules for the Education System, Order No. 974 of the Ministry of Education and Science, 15.08.2016**, Section IV, points 2, 7 to 10, and 12: heating checked before each season and recorded; firing ended two hours before sleep where people stay round the clock; no temporary stoves; boiler rooms run by trained people, never unattended; no solid fuel fired in a strong wind. Stove heating is not forbidden. Whether the site falls under these rules depends on its legal form.
- **Sanitary Regulation for General Secondary Education, Order No. 2205 of the Ministry of Health, 25.09.2020**: for schools, stove heating only in single-storey rural buildings, no iron stoves, and the air tested for carbon monoxide twice in each heating season. Whether it binds this site depends on its legal form.
- **ДСанПіН 5.5.5.23-99**, on children's recreation establishments: living rooms at least 18 °C; its approving act differs between sources.
- **Forest Code, No. 3852-XII of 21.01.1994**, amended last by Law No. 4017-IX of 10.10.2024: Art. 66, free gathering for one's own use, which does not name wood; Art. 69, special use of the forest needs a free felling ticket, issued by the owner or permanent user for fellings other than main felling; Art. 70, no harvesting in primeval or natural forests.
- **Cabinet Resolution No. 761 of 23.05.2007**, the procedure for special use of forest resources, amended last by No. 1154 of 17.09.2026: standing deadwood is kept. **Cabinet Resolution No. 929 of 22.10.2008**, felling rules for the mountain forests of the Carpathians, with a ticket and a technological map before work. **Cabinet Resolution No. 555 of 27.07.1995**, sanitary rules in forests. **Law No. 1436-III of 10.02.2000**, no clear felling in the Carpathian fir-beech forests above 1,100 m. Whether an owner may take brushwood or fallen wood from their own forest without a ticket was not found; the electronic ticket after the experiment that ended on 01.06.2025 was not verified.
- **ДБН В.2.5-64:2012**, with Amendment 1: hot water at 55 °C or more in systems where it applies.

- **Not reached:** ДБН В.2.5-67:2013, the clauses on chimneys and stoves; ДБН В.2.5-77:2014, the boiler output above which a boiler room is required.

The written enquiries: to the regional emergency service, the clearances, the woodshed's distance, the boiler room, and how the two-hour rule applies to a boiler with a buffer; to the regional forestry authority, whether the household's woodland is forest-fund land, and what may be felled for firewood and on what ticket; to the sanitary service, which rules bind the site's heating, once its legal form is settled.

11. Adaptation notes

The design is a seed, not a copy: a different place grows a different form from it.

- **A colder valley:** a larger buffer for the longer night; more wood, bought earlier; the expansion vessel and its pipes insulated.
 - **A milder valley:** a smaller buffer; the masonry heater may carry a building alone.
 - **A smaller group:** one masonry heater and a water heater may carry it; no boiler.
 - **A larger group:** a boiler room, if the rules ask it above a boiler's output.
 - **No woodland:** every cubic metre bought; nothing changes but the price.
 - **A stream:** the micro-hydro module's surplus into the buffer.
 - **Buildings that allow no gravity loop:** pumped radiators, with a gravity heat leak at the boiler.
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12. Provenance and licence

- **Written by:** Michel Garand. [set — others as they choose to be named]
- **From whose knowledge:** the energy working paper, EN v0.1, and the sanitation working paper, EN v0.2; ДСТУ-Н Б В.1.1-27:2010; ДБН В.2.6-31:2021; Reonic, 2026; LWF Merkblatt 12, 2024; LWF and TFZ on drying; ГОСТ 3243-88; Kiwa for BEIS, 2019; the 1. BImSchV, as given by BDH, 2011; the Bavarian LfU; Koop, haustec, 2022; Fröling's data sheet; German Wikipedia on gravity heating; Klimosz and Ukrainian installer guides; PN-B-02413:1991, as taught at Wrocław; the Solid Fuel Association; Watts; SNiP 2.04.01-85 and DIN 4708, on hot water; Carbon Trust for BEIS, 2020; energie-experten, 2020; Burch and Salasovich, NREL, 2005; Solarserver, 2025; the Masonry Heater Association, 1998; US EPA, Burn Wise; the Ukrainian instruments in Section 10, as read on secondary copies.
- **Traditional knowledge:** none taken. The literature's account of the stove, the cooking ledge, and the firewood shed is noted in the pattern line, for the elders to confirm, correct, or withhold.
- **Text:** CC BY-SA 4.0

- **Drawings and designs:** [hardware licence — open decision]
 - **Measurement data:** none yet
 - **Gate 1:** household decision, 2026-10-03, recorded by Michel Garand — v0.1 may be published as it stands, at L0 as a marked draft, in English, Ukrainian, and German
 - **Wartime review:** 2026-10-03, Michel Garand — v0.1 read against compendium Section 2; nothing places the site; the shelter, the safety link, the battery, the generator, and the fuel store in standing format; accepted
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13. What to check

- **The buildings** — surveyed; each one's heat load at -20°C , by EN 12831 or an engineer.
 - **The boiler and the buffer** — sized from the loads and the night.
 - **The gravity loop** — whether each building allows it.
 - **The thermal discharge valve** — whether 1 to 2 bar is enough.
 - **The hot water** — hot_water_per_person and the cold water's temperature, from a meter and a thermometer.
 - **The collectors** — their area and the roof they share with the array.
 - **The firewood** — a dated local price by the складометр; the wood's moisture on delivery.
 - **The woodland** — its category and what may be felled, from the enquiry.
 - **The elders** — the four questions in the pattern line.
 - **The enquiries** — clearances, woodshed, boiler room, and the two-hour rule.
 - **Quotations** — every line in Section 6, and the unpriced ones.
 - **The climate zone** — on the official map of ДБН В.2.6-31:2021.
 - **The Ukrainian row** — every instrument read and signed.
 - **From L2, once built:** build times and out-of-order consequences in Section 5; the maintenance record in Section 7; early signs and repair costs in Section 8; Section 9.
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